



Effects of various concentrations and volumes of potassium sorbate on colostrum bacterial contamination over time at ambient and refrigeration temperatures

Katharine S. Denholm,*  Maria Carvalho,  Alexandra Haggerty, and Rheinallt Jones 

School of Biodiversity, One Health and Veterinary Medicine, University of Glasgow, 464 Bearsden Road, Bearsden, Glasgow, G61 1QH, Scotland, United Kingdom

ABSTRACT

Preservation and storage of colostrum on farm is challenging, as cold temperature storage facilities (refrigerators and freezers) often cannot accommodate the volume of colostrum required to feed calves. Potassium sorbate preservative (PSP) is a readily available food-grade preservative used in wine and cheese production, shown to be very effective at preserving colostrum at both refrigeration and ambient temperatures. Aims of the current work were to determine the optimal concentration and inclusion volume rate (IVR) of PSP to preserve bovine colostrum for feeding to neonatal calves. Twenty individual cow point-of-feeding colostrum samples were collected and frozen at -20°C for 4 to 6 wk before testing (purposely selected). Three PSP concentrations were created (25.4%, 32.2%, and 39.1%) by mixing potassium sorbate crystals with 100 mL of distilled water. For each of these concentrations, either 50 μL (1% by volume) or 25 μL (0.5% by volume) were added to each 5-mL aliquot of colostrum to create 2 different IVR. For each concentration and IVR, one aliquot was stored at ambient temperature (AT; temperature monitored in the laboratory between 23.1°C and 24.6°C) and one aliquot was refrigerated (refrigeration temperature [RF]; temperature monitored in refrigerators between 3.1°C and 4.2°C). Baseline bacteria counts were also measured for each colostrum sample. Aliquots were tested for degree of bacterial contamination at 24, 72, and 168 h. Mixed linear regression models were constructed using total coliform count (TCC) and total bacteria count (TBC) as outcomes of interest, with concentration of potassium sorbate in distilled water (30 g/100 mL, 40 g/100 mL, or 50 g/100 mL), IVR (1% inclusion volume and 0.5% inclusion volume) and temperature (RF = 3.1°C – 4.2°C or AT = 23.1°C – 24.6°C) as predictors of interest. Sur-

vival analysis (with sample as a random effect term) was also used to account for bacteria counts too numerous to count (TNTC, assigned as 10,000,000 cfu/mL for TBC and 11,000 cfu/mL for TCC as these exceeded the highest measures). Bacteria TNTC was analyzed as a “failure event” and the time frame at which these failures occurred was analyzed. There was a significant temperature by IVR interaction for the TCC outcome. The RF conditions were more effective than AT in preventing bacterial proliferation at 24 and 72 h. Survival models showed a lower hazard of high contamination at IVR of 1% and concentration of 50 g/100 mL. In all cases, 50 g/100 mL concentration was more effective than concentrations of 30 g/100 mL and 40 g/100 mL or no preservative, and 1% IVR was more effective than 0.5% IVR or no preservative at preventing bacterial proliferation at 24 and 72 h. A large number of samples in this work were initially contaminated with total bacterial species and this may have biased results; however, this is representative of the current situation on most dairy farms.

Key words: calves, colostrum, preservation

INTRODUCTION

Bacterial contamination of bovine colostrum is a problem on a national (Hyde et al., 2020; Haggerty et al., 2021; Anderson et al., 2024) and international scale (Morrill et al., 2012; Phipps et al., 2016; Godden et al., 2025).

Bacteria may contaminate colostrum at harvest, storage, or feeding (Hyde et al., 2020) through inadequate hygiene practices, including inadequate udder preparation and inadequate cleaning of colostrum storage and feeding equipment (Hyde et al., 2020; Haggerty et al., 2025). Heat treatment of colostrum before feeding may limit transfer of bacteria (Godden et al., 2012; Mann et al., 2020), but pasteurization units are not commonplace on UK dairy farms (Denholm et al., 2023). Coliform bacteria are of particular importance, as they can disrupt the transfer of passive immunity from dam to neonate (James et al., 1981; Johnson et al., 2007; God-

Received September 18, 2025.

Accepted December 8, 2025.

*Corresponding author: katie.denholm@glasgow.ac.uk

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

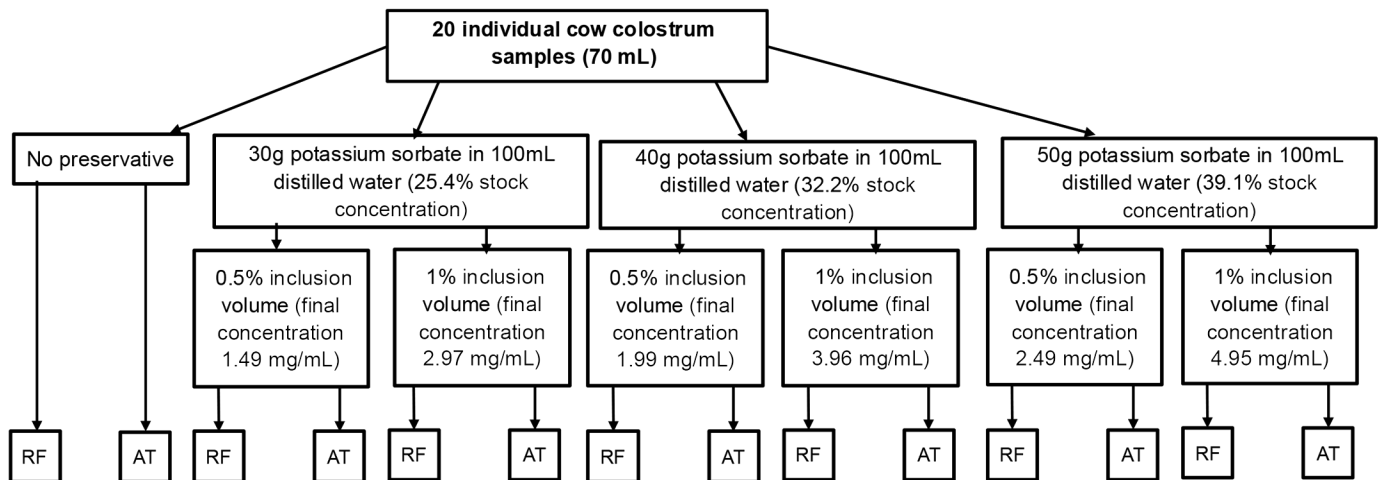


Figure 1. Flowchart to show the various aliquots created from 20 individual cow colostrum samples stored up to 168 h and measured for total coliform and total bacteria counts (cfu/mL) using Petrifilms. AT = ambient temperature, measured in the laboratory between 23.1°C and 24.6°C; RF = refrigeration temperature, measured between 3.1°C and 4.2°C.

den et al., 2012). Maximum bacterial counts of 10,000 cfu/mL and 100,000 cfu/mL have been proposed for total coliform count (TCC) and total bacteria count (TBC), respectively, although published calf health related associations with these thresholds are limited (Mellado et al., 2017).

Surplus first milking colostrum or transition milk may be stored for later feeding to calves. In some cases, stored colostrum from other dams may be required to feed to neonatal calves where colostrum supply is limited, or to prevent disease transmission from dam to calf (Denholm et al., 2017; Godden et al., 2025). An increasing body of literature suggests that there are health and growth benefits of feeding colostrum or transition milk (milking 2–6 after calving; Godden, 2008; Van Soest et al., 2022) to calves beyond the first 24 h of life (Conneely et al., 2014; Chamorro et al., 2017; Kargar et al., 2020). In light of this research, effective methods for preserving and storing colostrum for extended feeding are of increasing interest.

Preservation and storage of colostrum on farm is challenging, as cold temperature storage facilities (refrigerators and freezers) often cannot accommodate the volume of first milking colostrum required to feed numerous neonatal calves (4–6 L for a 40-kg Holstein-Friesian dairy calf; Godden et al., 2019). Furthermore, temperatures are seldom monitored in these devices (Haggerty et al., 2021). It has been demonstrated that temperatures of 4°C and –20°C are favorable for storage of colostrum for 2 to 3 d (Cummins et al., 2017) and 6 to 12 mo (Stewart et al., 2005), respectively. More recent work suggests that maximum storage time for stored colostrum frozen at –20°C is 32 wk to prevent changes to composition (Westhoff and Mann 2025).

Acid preservatives (such as citric, propionic, and formic acids) are variably efficacious at preserving colostrum quality and may affect palatability (Denholm 2022). Unlike acids, which act to extend the lag phase of bacterial growth by lowering colostrum pH, potassium sorbate preservative (PSP) prevents bacterial proliferation by altering cell membranes, inhibiting cell transport systems and enzyme function (Sofos et al., 1986). Potassium sorbate is a readily available food-grade preservative used in wine and cheese production and has been shown to be very effective at preserving colostrum (Denholm et al., 2017; Godden et al., 2025). Early work by Stewart et al., (2005) showed that PSP may be used to preserve colostrum for up to 96 h together with refrigeration at 4°C. More recent work has shown that PSP may be used effectively at ambient temperatures (without the need for refrigeration) for 96 h (Godden et al., 2025) up to 7 d (Denholm et al., 2017) or even up to 14 d (Cuttance et al., 2025), depending on initial bacterial contamination. Some of this work has used pooled colostrum samples from seasonal calving systems (Denholm et al., 2017; Cuttance et al., 2025), and some used individual cow samples (Godden et al., 2025). At present, the optimal concentration for potassium sorbate is unclear; with 50% concentration (Denholm et al., 2017; Cuttance et al., 2025) and a 38% concentration stabilized liquid solutions previously reported (Godden et al., 2025). In addition, the optimal inclusion volume rate (IVR) for potassium sorbate is unclear with works advocating for a 1% by volume dilution (Denholm et al., 2017; Cuttance et al., 2025) and a 0.3% to 0.5% by volume dilution (Stewart et al., 2005; Godden et al., 2020, 2025).

Table 1. Baseline total bacteria count and total coliform count (cfu/mL) measurements for 20 previously frozen individual cow colostrum samples collected from a single Scottish dairy farm¹

Sample	Total bacteria counts (cfu/mL)	Exceeded industry total bacteria count threshold (100,000 cfu/mL)	Total coliform count (cfu/mL)	Exceeded industry total coliform count threshold (10,000 cfu/mL)
1	462,500	Yes	708	No
2	1,467	No	708	No
3	2,025	No	667	No
4	780,000	Yes	500	No
5	433,333	Yes	833	No
6	16,250	No	0	No
7	29,250	No	375	No
8	628,333	Yes	0	No
9	16,250	No	125	No
10	866,667	Yes	75	No
11	208	No	3,542	No
12	1,256,667	Yes	250	No
13	>10,000,000	Yes	1,500	No
14	625	No	1,208	No
15	715,000	Yes	1,250	No
16	671,667	Yes	833	No
17	52,000	No	125	No
18	8,385,000	Yes	500	No
19	2,417	No	0	No
20	>10,000,000	Yes	>11,000	Yes

¹Bacterial counts were measured using Petrifilms.

The aims of this work were to determine the optimal concentration and IVR of PSP to preserve bovine colostrum for feeding to neonatal calves. The hypothesis was that a concentration of 50 g/100 mL of water added to colostrum at an IVR of 1% would be more effective at preventing bacterial proliferation than lower concentration of 30 g/100 mL and 40 g/100 mL of water and lower inclusion volume rates of 0.5%. It was also hypothesized that refrigeration temperatures would be more effective than ambient temperatures to preserve colostrum.

MATERIALS AND METHODS

Sample Preparation

Twenty individual cow point-of-feeding (collected immediately before feeding to the calf) colostrum samples were collected from a single dairy farm in Scotland and frozen at -20°C for 4 to 6 wk before testing. Our sample size estimation was based on prior data of similar studied by Godden et al. (2025; 15 individual cow colostrum samples aliquoted to assess the effects of 38% stabilized potassium sorbate at refrigeration and room temperature) and time constraints and practicalities of preparing multiple aliquots in the laboratory. Samples were analyzed under University of Glasgow ethics (ref EA22/25). Samples were purposively selected from stored samples during the study period (May to July 2025). Samples were thawed at room temperature before testing for approximately 4 h. Fourteen aliquots of each individual colostrum sample were created as shown in Figure 1.

Various concentrations of potassium sorbate were created by mixing 30, 40, and 50 g of food-grade potassium sorbate crystals (Intra Laboratories, Plymouth, UK) with 100 mL of distilled water:

- 30 g of potassium sorbate in 100 mL of distilled water mixed up to 118 mL, creating a 25.4% concentration,
- 40 g of potassium sorbate in 100 mL of distilled water mixed up to 124 mL, creating a 32.2% concentration, and
- 50 g of potassium sorbate in 100 mL of distilled water mixed up to 128 mL, creating a 39.1% concentration.

For each of these concentrations 50 μL (1% by volume) and 25 μL (0.5% by volume) were added to each 5-mL aliquot of colostrum to create 2 different IVR for each concentration. For each concentration and IVR, one aliquot was stored at ambient temperature (AT; temperature monitored in the laboratory between 23.1°C and 24.6°C) and one aliquot was stored at refrigeration temperatures (RF; temperature monitored in refrigerators between 3.1°C and 4.2°C). Aliquots were tested for bacterial contamination at 24, 72, and 168 h.

All experimental samples were tested in a single replicate. A small number of duplicate samples were performed and coefficient of variation calculated (duplicate counts $n = 16$ for TBC and $n = 8$ for TCC). There was a low coefficient of variation for each of the Petrifilms (1.3% for TCC and 0.9% for TBC; 3M, St. Paul, MN). Previous work also demonstrated a low coefficient of variation of 3.4% for TBC Petrifilms (Senyk et al., 1987).

Table 2. Descriptive statistics for total coliform and total bacteria count (cfu/mL) from 19 individual cow colostrum samples from a single Scottish dairy farm measured by Petrifilms over 168 h under various storage conditions

Item	Measure	24 h	72 h	168 h
Total coliform count (cfu/mL)	Median	1.33×10^3	1.42×10^3	8.33×10^2
	Minimum	0	0	0
	Maximum	1.10×10^4	1.10×10^4	1.10×10^4
	Interquartile range	5.44×10^3	1.06×10^4	2.25×10^3
Total bacteria count (cfu/mL)	Median	1.00×10^7	1.00×10^7	1.00×10^7
	Minimum	0	475.00	50.00
	Maximum	1.00×10^7	1.00×10^7	1.00×10^7
	Interquartile range	9.30×10^6	7.79×10^6	0

Bacteria Testing: Main Outcome of Interest

Baseline bacterial contamination was measured using coliform (TCC) and total aerobic bacteria (TBC) count Petrifilms for each of the 20 previously frozen individual cow colostrum samples (Table 1). Colostrum samples were prepared according to the protocols published by Moore and Sisco (2015). Briefly, 1:125 dilutions were prepared (using PBS) for the TCC Petrifilms and 1:3,250 dilutions for the TBC Petrifilms, and 1 mL of each of these dilutions was inoculated onto the films. The TCC Petrifilms were incubated at $35^\circ\text{C} \pm 1^\circ\text{C}$ for 24 h and TBC Petrifilms were incubated at $32^\circ\text{C} \pm 1^\circ\text{C}$ for 48 h. Colonies were counted using a backlit colony counter (Stuart Scientific SC5 Colony Counter, Merck). Each aliquot was measured using TCC and TBC Petrifilms at 24, 72, and 168 h (hereafter referred to as d 1, 3, and 7, respectively).

Statistical Analysis

Data were analyzed using Microsoft Excel (Office 16), Stata (StataCorp LLC version 18), and R (version 4.5.1, <https://www.r-project.org/>).

Mixed linear (repeated measures) regression models were constructed using TCC and TBC as outcomes of interest, with concentration (30 g/100 mL; 40 g/100 mL, or 50 g/100 mL), IVR (1% inclusion volume and 0.5% inclusion volume), temperature (RF = 3.1°C – 4.2°C or AT = 23.1°C – 24.6°C) and time (24, 72, and 168 h) as predictors of interest. One sample was removed from further analysis as both baseline TBC and TCC measures were too numerous to count (TNTC). Sample ID (n = 1 to 19) was included as a random effect term. Bacteria counts were log-transformed as they were right skewed and violated the assumption of normality of the mixed linear effects models. Because many of the initial TBC measures were TNTC (n = 11/20, see Table 1), samples were stratified such that only those samples (n = 9) that were under the threshold of 100,000 cfu/mL at initial baseline testing

were included in the multivariable linear models for the log-transformed TBC outcomes and the multilevel survival models. The highly contaminated samples ($\geq 100,000$ cfu/mL) were analyzed separately and the multivariable linear modeling for these (n = 10) is presented separately.

Each predictor variable was tested in turn at the univariable level and interaction between the predictors of interest was also explored, including between temperature and concentration; temperature and IVR, and concentration and IVR. All variables of interest were included in multivariable linear mixed effects models. The assumption of linearity of continuous variables in the linear models was assessed by plotting the outcome against each variable. Model diagnostics for the final models included plotting residuals, which were found to lie within 2 SD of the mean.

Survival analysis (with sample as a frailty term) was also used to account for bacteria counts TNTC (assigned as 10,000,000 cfu/mL for TBC and 11,000 cfu/mL for TCC, as these exceeded the highest measures on the Petrifilms). Counts measured as TNTC were analyzed as a failure event, and the time frame at which these failures occurred was analyzed. Cox proportional hazards models were constructed, using robust SE to account for clustering in the data. A sample frailty term was included in the Cox proportional hazards models. The ‘stcurve’ function in Stata was used to estimate the cumulative hazards function and no violation of model assumptions was observed.

RESULTS

Table 1 shows the baseline bacteria measurements for the 20 samples used in this work (median TBC = 4.5×10^5 cfu/mL, interquartile range [IQR] = 7.9×10^5 cfu/mL and median TCC = 5.8×10^2 cfu/mL, IQR = 8.0×10^2 cfu/mL). Descriptive statistics for the 19 samples used in this analysis at 24, 72, and 168 h are presented in Table 2 and Figure 2. A large number of aliquots were TNTC throughout the experiment. For

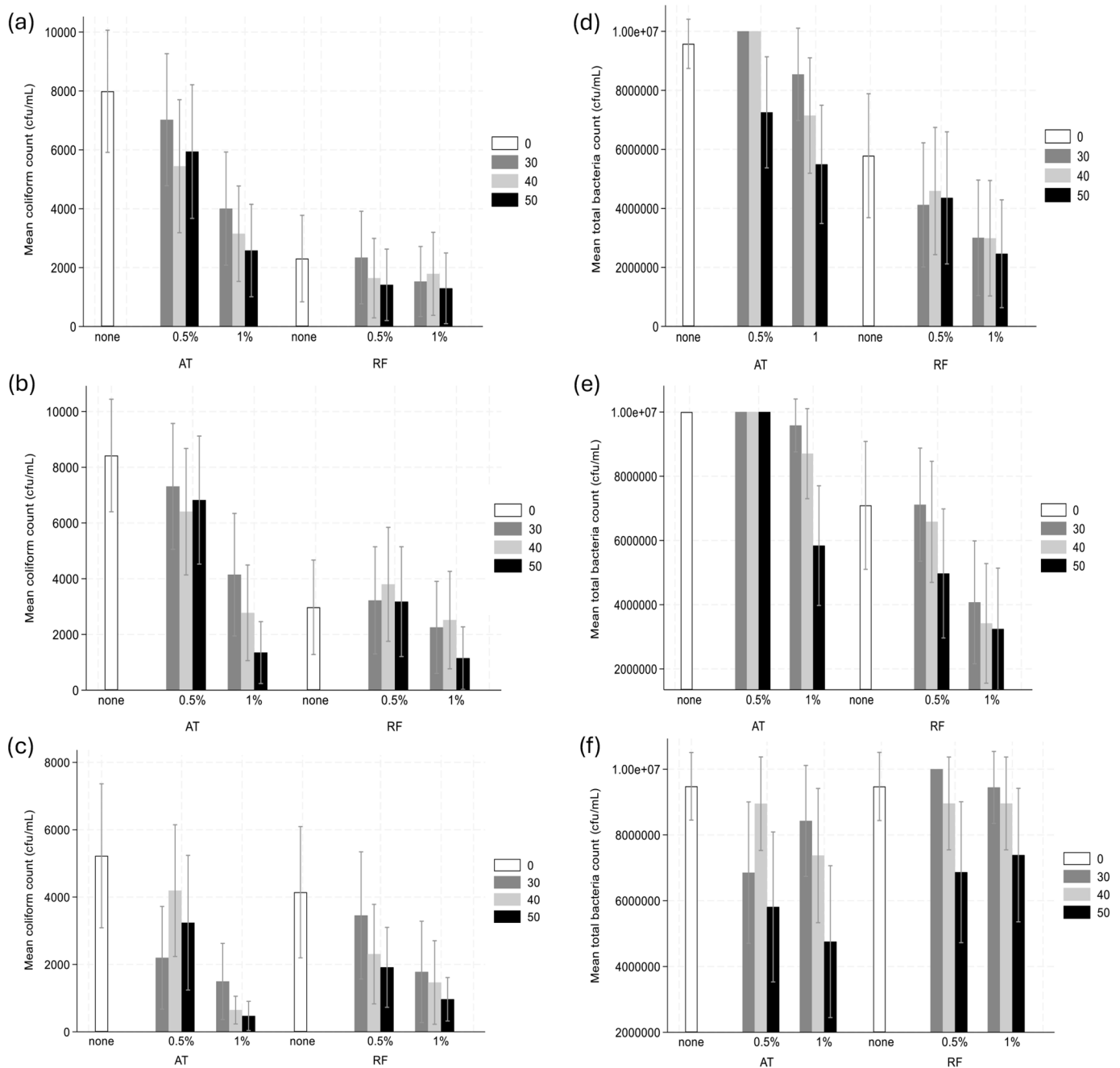


Figure 2. Descriptive charts to show total coliform counts (cfu/mL) at (a) 24 h, (b) 72 h, and (c) 168 h and total bacteria counts (cfu/mL) at (d) 24 h, (e) 72 h, and (f) 168 h for 19 individual cow colostrum samples preserved using various concentrations (30 = 30 g of potassium sorbate in 100 mL of distilled water, 25.4% concentration; 40 = 40 g of potassium sorbate in 100 mL of distilled water, 32.2% concentration; and 50 = 50 g of potassium sorbate in 100 mL of distilled water, 39.1% concentration) and by inclusion volume percentages of potassium sorbate at ambient temperatures (AT, 23.1°C–24.6°C) and refrigeration temperatures (RT, 3.1°C–4.2°C). 0.5% indicates that 25 μ L of each dilution rate was added to each 5-mL aliquot; 1% indicates that 50 μ L of each dilution rate was added to each 5-mL aliquot. Bacteria counts were measured using Petrifilms, and where bacteria were too numerous to count (TNTC), a value of 11,000 cfu/mL was assigned to the measurements for coliform counts, and 1×10^7 cfu/mL was assigned to the measurements for total bacteria counts. Error bars indicate CI.

TCC at 24 h, 58/266 (21.8%) aliquots were TNTC; at 72 h, 79/266 (29.7%) aliquots were TNTC; and at 168 h, 34/266 aliquots (12.8%) were TNTC. For TBC,

154/266 (57.9%) aliquots were TNTC; at 72 h, 179/266 (67.3%) aliquots were TNTC, and at 168 h, 214/266 (80.5%) aliquots were TNTC.

Table 3. Final mixed linear effects model for log-transformed total coliform count (cfu/mL) for 19 individual cow colostrum samples preserved using various concentrations (30 g/100 mL, 40 g/100 mL, and 50 g/100 mL) and various volumes (0.5% and 1%) of potassium sorbate preservative¹

Variable	Category	Coefficient	SE	P-value	95% CI
Constant		3.25	0.21	<0.001	2.83 to 3.66
Baseline coliform count (cfu/mL)		5.4×10^{-4}	1.7×10^{-4}	0.001	2.1×10^{-4} to 8.7×10^{-4}
Volume of potassium sorbate (%)	0.5	-0.65	0.13	<0.001	-0.91 to -0.38
	1	-1.35	0.13	<0.001	-1.62 to -1.09
Concentration of potassium sorbate ² (g/100 mL distilled water)	30	-1.00	0.14	<0.001	-1.27 to -0.73
	40	-1.14	0.14	<0.001	-1.41 to -0.88
	50	-1.35	0.14	<0.001	-1.62 to -1.09
Storage temperature (°C)	Refrigeration (3°C–4°C)	-0.59	0.16	<0.001	-0.90 to -0.28
Temperature-by-volume interaction	Refrigeration by 0.5%	0.21	0.18	0.24	-0.14 to 0.57
	Refrigeration by 1%	0.62	0.18	<0.001	0.27 to 0.98
Timing	72 h	0.17	0.07	0.02	0.03 to 0.31
	168 h	-0.13	0.07	0.06	-0.27 to 0.01

¹Samples were stored at ambient (23.1°C–24.6°C) and refrigeration (3.1°C–4.2°C) temperatures for 24, 72, and 168 h and tested for bacterial contamination using Petrifilms. Random effect terms: sample number (1 to 19) Estimate = 0.69, SE = 0.04, R² = 0.25.

²30 = 30 g of potassium sorbate in 100 mL of distilled water, 25.4% concentration; 40 = 40 g of potassium sorbate in 100 mL of distilled water, 32.2% concentration; and 50 = 50 g of potassium sorbate in 100 mL of distilled water, 39.1% concentration.

Tables 3 and 4 show the final multivariable models for the log-transformed TCC (n = 19 samples) and TBC measures (n = 9 samples, <100,000 cfu/mL), respectively. In addition, highly contaminated colostrum samples ($\geq 100,000$ cfu/mL TBC) were stratified and analyzed separately, and these results are presented in Table 5. In all cases, 50 g/100 mL concentration was more effective than concentration of 30 g/100 mL and 40 g/100 mL or no preservative, and 1% IVR was more effective than 0.5% IVR or no preservative at preventing bacterial proliferation at 24 and 72 h. Baseline bacteria measures were retained in the models as significant ($P = 0.001$) predictors for TCC, but had no significant effect ($P = 0.46$) on TBC outcomes (however, they were retained in the final models). There was a significant ($P < 0.001$) temperature-by-volume interaction for the TCC outcome. In all cases, RF was more effective than AT in preventing bacterial proliferation at 24 and 72 h.

Table 6 shows the Cox proportional hazards models for TCC and TBC measures. As in the linear mixed effect models, baseline measures (as shown in Table 1) were included in the TCC outcome models and were forced into the TBC models.

DISCUSSION

The objective of this study was to determine the optimal concentration and IVR of PSP added to first milking colostrum for feeding to newborn calves. In the current work, 50 g of potassium sorbate in 100 mL of distilled water (39.1% solution) was more effective at preventing bacterial proliferation (TCC and TBC) than the lower concentration of 30 or 40 g in 100 mL of distilled water in support of the study hypothesis. Inclusion volume rates of 1% were more effective than IVR of 0.5% and RF

temperatures (3.1°C–4.2°C) were more effective at preventing bacterial proliferation than AT (23.1°C–24.6°C), again in support of the study hypothesis.

Many (n = 11/20, 55%) of the individual cow first milking colostrum samples used in this study initially exceeded the TBC threshold guidelines of 100,000 cfu/mL (McGuirk and Collins, 2004; Godden et al., 2019); however, only one sample exceeded TCC threshold guidelines (of 10,000 cfu/mL, see Table 1). Some of the prolific mesophilic aerobic bacteria measured by the TBC Petrifilms may be beneficial species, as TBC is an all-encompassing measure of all mesophilic species (both potentially beneficial or detrimental to the calf). In fact, it has been demonstrated that calves deprived of early colostrum feeding are less likely to have beneficial bacterial species such as *Bifidobacterium* spp. and *Lactobacillus* spp. in their colon, which may be a marker of future immune response and health (Frizzo et al., 2010; Fischer et al., 2018). As such, coliform counts may be a more accurate representation of colostrum contamination (Cuttance et al., 2025). High levels of bacteria in colostrum, and particularly coliform bacteria, may bind free Ig in the gut lumen or directly block uptake and transport of Ig molecules across intestinal epithelial cells, thus interfering with passive transfer (James et al., 1981; Godden et al., 2019). Indeed, coliform species have been used as indicator organisms for bacterial contamination in dairy foodstuffs (Martin et al., 2016). In microbiological testing, an indicator organism is defined as a marker that reflects the general microbiological condition of a food or environment (Chapin et al., 2014), and diverse coliform species have been found to come from both fecal and environmental sources (Martin et al., 2016).

Nevertheless, first milking colostrum has been shown to be consistently highly contaminated with myriad bac-

Table 4. Final mixed linear effects model (with timing and sample number as random effect terms) for log-transformed total bacteria counts (cfu/mL) for 9 individual cow colostrum samples (<100,000 cfu/mL) preserved using various concentrations (30 g/100 mL, 40 g/100 mL, and 50 g/100 mL) and various volumes (0.5% and 1%) of potassium sorbate preservative¹

Variable	Category	Coefficient	SE	P-value	95% CI
Constant		6.84	0.32	<0.001	6.22 to 7.46
Baseline total bacteria count (cfu/mL)		-1.46×10^{-4}	1.99×10^{-4}	0.46	-5.35×10^{-4} to 2.44×10^{-4}
Volume of potassium sorbate (%)	0.5%	-0.31	0.18	0.09	-0.67 to 0.05
	1%	-0.66	0.18	<0.001	-1.02 to -0.30
Concentration of potassium sorbate ² (g/100 mL distilled water)	30	-0.42	0.18	0.02	-0.78 to -0.06
	40	-0.48	0.18	0.009	-0.84 to -0.12
	50	-0.66	0.18	<0.001	-1.02 to -0.30
Storage temperature (°C)	Refrigeration (3°C–4°C)	-0.59	0.11	<0.001	-0.80 to -0.38
Timing	72 h	0.51	1.34	<0.001	0.24 to 0.77
	168 h	0.28	1.34	0.04	-0.16 to 0.54

¹Samples were stored at ambient (23.1°C–24.6°C) and refrigeration (3.1°C–4.2°C) temperatures for 24, 72, and 168 h and tested for bacterial contamination using Petrifilms. Random effect terms: sample number (1 to 19) Estimate = 0.12, SE = 0.26, $R^2 = 0.12$.

²30 = 30 g of potassium sorbate in 100 mL of distilled water, 25.4% concentration; 40 = 40 g of potassium sorbate in 100 mL of distilled water, 32.2% concentration; and 50 = 50 g of potassium sorbate in 100 mL of distilled water, 39.1% concentration.

teria due to contamination from cows' udders, storage buckets, or feeders (Hyde et al., 2020; Haggerty et al., 2025), and more consistent efforts to minimize bacterial contamination before storage are warranted to harness the benefits of PSP for colostrum (Denholm et al., 2025).

Temperature

Bacterial species proliferate most rapidly in the first 6 h of preservation, particularly at warmer temperatures of 20°C compared with cooler temperature of 4°C and 13°C (Cummins et al., 2016). The same work observed that when the TBC of colostrum stored at 4°C exceeded 1,000,000 cfu/mL, bacterial growth rate declined (Cummins et al., 2016); this is relevant to the current work, as many of the aliquots were TNTC throughout the experiment, and a similar decline was observed in some samples.

Initial work on the use of PSP to preserve colostrum showed that it is preferentially used at refrigeration temperatures (4°C; Stewart et al., 2005; Bey et al., 2007); although more recent work demonstrated its effectiveness to maintain IgG concentration and minimize bacterial proliferation even at ambient temperatures (Denholm et al., 2017; Godden et al., 2025). Potassium sorbate may be particularly useful on farms where cold temperature storage facilities are limited (none or limited capacity refrigerators and freezers). It is important to note that on many farms, even where cold temperature storage is employed, refrigeration and freezing temperatures are seldom monitored (Haggerty et al., 2021), such that cold temperature storage conditions may be suboptimal for colostrum. It has been suggested that 4°C for refrigeration and -20°C for freezing may be suitable (Stewart et al., 2005; Denholm 2022). In the current study, RF tempera-

tures preserved colostrum more effectively than AT although concentration of 50 g/100 mL and IVR of 1% performed well at both temperatures.

Concentration

Although PSP is safe for animal consumption (EFSA Panel on FEEDAP, 2015) it is essential to minimize the amount of this chemical ingested by calves, to prevent potential toxicity (European Food Safety Authority safe concentration 9 g/kg BW complete feed for young ruminants, refer to Supplemental Table S1 [see Notes] for acceptable doses of potassium sorbate in all species). Other published work asserts that typical PSP levels used in many human food products (250–1,000 ppm) are considered completely safe (McGuirk and Collins 2004). Hence, the purpose of this work was to establish the lowest concentration of PSP that could be used to effectively preserve colostrum. There is some confusion in the published literature on the nomenclature of the various concentrations, because in some work, 50 g/100 mL of distilled water is referred to as a “50% solution” (Denholm et al., 2017; Cuttance et al., 2025), when in fact 50 g of PSP in 100 mL of distilled water dissolves to 128 mL, resulting in 39.1% concentration. In the current work, lower concentrations were also tested (40 g/100 mL concentration = 32.2% solution and 30 g/100 mL concentration = 25.4% solution) and did not perform as well as the 50 g/100 mL concentration. Should a 40 kg dairy calf consume 10% to 12% of its BW (Godden et al., 2019) in first milking colostrum preserved with PSP at concentration 50 g/100 mL and at IVR 1%, that animal would potentially ingest 20 to 30 g of potassium sorbate (5 g potassium sorbate in 10 mL of solution and 10 mL of solution added to each liter of colostrum), which is well within the safety margins.

Table 5. Final mixed linear effects model (with timing and sample number as random effect terms) for log-transformed total bacteria counts (cfu/mL) for 10 highly contaminated ($\geq 100,000$ cfu/mL total bacteria count) individual cow colostrum samples preserved using various concentrations (30 g/100 mL, 40 g/100 mL, and 50 g/100 mL) and various volumes (0.5% and 1%) of potassium sorbate preservative¹

Variable	Category	Coefficient	SE	P-value	95% CI
Constant		6.90	0.16	<0.001	6.59 to 7.21
Baseline total bacteria count (cfu/mL)		1.41×10^{-8}	2.72×10^{-8}	0.52	-3.92×10^{-8} to 6.73×10^{-8}
Volume of potassium sorbate (%)	0.5%	-0.58	0.16	<0.001	-0.88 to -0.27
	1%	-0.66	0.18	<0.001	-1.17 to -0.55
Concentration of potassium sorbate ² (g/100 mL distilled water)	30	-0.28	0.16	0.08	-0.59 to 0.03
	40	-0.31	0.16	0.05	-0.62 to 0.01
	50	-0.86	0.16	<0.001	-1.17 to -0.55
Storage temperature (°C)	Refrigeration (3°C–4°C)	-0.26	0.09	0.005	-0.44 to -0.08
Timing	72 h	0.07	0.11	0.56	-0.16 to 0.29
	168 h	-0.12	0.11	0.27	-0.35 to 0.10

¹Samples were stored at ambient (23.1°C–24.6°C) and refrigeration (3.1°C–4.2°C) temperatures for 24, 72, and 168 h and tested for bacterial contamination using Petrifilms. Random effect terms: sample number (1 to 19) Estimate = 0.15, SE = 0.07, $R^2 = 0.11$.

²30 = 30 g of potassium sorbate in 100 mL of distilled water, 25.4% concentration; 40 = 40 g of potassium sorbate in 100 mL of distilled water, 32.2% concentration; and 50 = 50 g of potassium sorbate in 100 mL of distilled water, 39.1% concentration.

Inclusion Volume Rates

Inclusion volume rates of 1% have been shown to be effective at preserving colostrum (Denholm et al., 2017; Cuttance et al., 2025; using a 39.1% concentration), as have lower IVR of 0.3% (Godden et al., 2025; using a 38% concentration) and 0.5% (Stewart et al., 2005; Bey et al. 2007; using a 38% solution and a 50% concentration, respectively). Godden et al. (2025) demonstrated the efficacy of 0.3% IVR to prevent proliferation of pathogenic species such as *Escherichia coli*, *Staphylococcus aureus*, and *Salmonella* spp. in colostrum. However, in the same study, PSP in bulk tank milk at a VR of 0.15% of a 38% stabilized solution of potassium sorbate was not as efficacious at preventing bacterial proliferation (Godden et al., 2025). The authors hypothesized that the volume added to the bulk milk was too low. In the current work 0.5% IVR was not as effective as 1% IVR. Further research is needed to ascertain optimal concentration and IVR for transition milk (milkings 2–6 after calving), which has also been shown to be highly contaminated (Denholm et al., 2025).

Potassium Sorbate Availability

In the United States, a commercial stabilized PSP is available, providing a 38% solution (DamFresh Solution, Bio Sciences Inc.). As mentioned, some recent work supports the use of this product in both first milking colostrum (IVR = 0.3%) and bulk tank milk (IVR = 0.15%) to prevent bacterial proliferation (Godden et al., 2025); however, this product is not available in Europe. In Australia, a liquid product containing 50% stabilized solution is available (although g/mL is undisclosed; Potassium Sorbate 50% Solution, The Vet

Group), and this product is recommended to be added at 1% by volume. In New Zealand, potassium sorbate crystals are available for agricultural use (VetPak, Te Awamutu, New Zealand) but farmers and veterinarians must dissolve these crystals in water before use and care must be taken to use uncontaminated water sources on farm to dissolve the product. To the authors' knowledge, no such feed grade agricultural product exists in the UK or Europe but potassium sorbate crystals are freely available to purchase for other purposes (wine and cheese production).

Study Limitations

A large number of samples analyzed in this work were initially contaminated with total mesophilic bacterial species, and this may have biased results; nonetheless, this is representative of the current situation on many dairy farms. Other work has demonstrated efficacy of PSP in clean samples (Godden et al., 2025), and in the current work the TBC samples were stratified such that the effect of PSP could be explored in samples with initial contamination <100,000 cfu/mL. Because many of the total bacterial counts were TNTC, analysis techniques, such as survival analysis, were selected to try to counteract this. In some cases, there were counterintuitive findings of reductions in bacterial counts, where proliferation was expected over time (particularly at 168 h), and this could have been due to inaccuracies in colony enumeration, depletion of colostrum nutrients for bacterial growth or preference of certain bacterial species for cooler temperature conditions. Frozen, rather than fresh colostrum samples were used in this work for logistical reasons, but this may also have biased results, as freezing may affect coliform viability (Alrabadi, 2015).

Table 6. Final multivariable Cox proportional hazards models for high total coliform ($\geq 11,000$ cfu/mL) and total bacteria counts ($\geq 1 \times 10^7$ cfu/mL) measured by Petrifilms ($n = 19$ coliform and $n = 9$ total bacteria) for individual cow colostrum samples preserved using various concentrations (30 g/100 mL, 40 g/100 mL, and 50 g/100 mL) and various volumes (0.5% and 1%) of potassium sorbate¹

Outcome	Variable	Category	Hazard ratio	Robust SE	P-value	95% CI
Total coliform count (cfu/mL)	Constant		0.0007	0.0003	<0.001	0.0003–0.001
	Concentration of potassium sorbate ² (g/100 mL distilled water)	30	2.71	1.04	0.009	1.28–5.74
		40	2.18	0.76	0.03	1.10–4.33
		50	1.86	0.70	0.10	0.89–3.89
	Volume (%)		0.32	0.11	0.001	0.16–0.64
	Temperature (°C)	Refrigeration (3.1°C–4.2°C)	0.37	0.14	0.007	0.18–0.76
Total bacteria count (cfu/mL)	Baseline measure (cfu/mL)		1.00	0.0002	<0.001	1.00–1.00
	Constant		0.0005	0.0002	<0.001	0.0003–0.001
	Concentration of potassium sorbate ² (g/100 mL distilled water)	30	1.09	0.09	0.28	0.93–1.28
		40	1.04	0.09	0.67	0.87–1.24
		50	0.94	0.09	0.50	0.78–1.13
	Volume (%)		0.77	0.10	0.04	0.60–0.98
	Temperature (°C)	Refrigeration (3.1°C–4.2°C)	0.78	0.07	0.005	0.65–0.92
	Baseline measure (cfu/mL)		1.00	1.03×10^{-4}	0.66	1.00–1.00

¹Samples were stored at ambient (23.1°C–24.6°C) and refrigeration (3.1°C–4.2°C) temperatures for 24, 72, and 168 h. Sample frailty term for total coliform counts hazard ratio = 0.35, robust SE = 0.22, sample frailty term for total bacteria counts hazard ratio = 0.017, robust SE = 0.015.

²30 = 30 g of potassium sorbate in 100 mL of distilled water, 25.4% concentration; 40 = 40 g of potassium sorbate in 100 mL of distilled water, 32.2% concentration; and 50 = 50 g of potassium sorbate in 100 mL of distilled water, 39.1% concentration.

In this work, Petrifilms were used to measure TCC and TBC rather than reference MacConkey agar and sheep blood agar plates, which may have resulted in less accurate bacterial measures. However, Petrifilms have been validated for use in colostrum in published literature (Morin et al., 2021; Anderson et al., 2024). Immunoglobulin G concentrations in colostrum were not measured as part of this study; however, it has been demonstrated that IgG concentrations are not detrimentally affected by PSP (Denholm et al., 2017; Godden et al., 2025) or temperature conditions (up to 60°C; Cummins et al., 2016). In addition, pH measures were beyond the scope of the current work, but pH has been shown to change (become more acidic) in highly contaminated colostrum samples (Cummins et al., 2016), and calves are likely to refuse acidic liquid feed (Collings et al., 2011; Hill et al., 2013).

CONCLUSIONS

In the current work, 50 g of potassium sorbate in 100 mL of distilled water (39.1% solution) was more effective at preventing bacterial proliferation (TCC and TBC) than the lower concentration of 30 g/100 mL and 40 g/100 mL of distilled water. Inclusion volume rates of 1% were more effective than IVR of 0.5% and RF temperatures (3.1°C–4.2°C) were more effective at preventing bacterial proliferation than AT (23.1°C–24.6°C).

NOTES

This study received no external funding. Thank you to Mike Denholm, Becky Orr, and Fiona Mackintosh for their assistance in the laboratory. Thank you to the farm staff who collected the colostrum samples used for this work. Supplemental material for this article is available at <https://doi.org/10.17632/75n5wyvrfe.1>. Samples were analyzed under University of Glasgow ethics (ref EA22/25). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AT = ambient temperature; IQR = interquartile range; IVR = inclusion volume rate; PSP = potassium sorbate; RF= refrigeration temperature; TBC = total bacteria count; TCC= total coliform count; TNTC= too numerous to count.

REFERENCES

- Alrabadi, N. I. 2015. The effect of freezing on different bacterial counts in raw milk. *Int. J. Biol.* 7:2013–2016. <https://doi.org/10.5539/ijb.v7n4p9>.
- Anderson, T., A. Haggerty, E. Silva, C. Mason, D. Bell, and K. S. Denholm. 2024. Validation of the diagnostic accuracy of the 3M Petrifilm coliform and aerobic count plates to measure colostrum bacterial contamination on Scottish dairy farms. *Vet. J.* 308:106230. <https://doi.org/10.1016/j.tvjl.2024.106230>.
- Bey, R., S. Godden, H. Lillegaard, S. Stewart, P. Rapnicki, J. Fetrow, and R. Farnsworth. 2007. Improving cleanliness and shelf-life of refrigerated colostrum using heat-treatment and chemical preservatives.

- University of Minnesota, University Digital Conservancy. Accessed Oct. 18, 2025. <https://hdl.handle.net/11299/109848>.
- Chamorro, M. F., N. Cernicchiaro, and D. M. Haines. 2017. Evaluation of the effects of colostrum replacer supplementation of the milk replacer ration on the occurrence of disease, antibiotic therapy, and performance of pre-weaned dairy calves. *J. Dairy Sci.* 100:1378–1387. <https://doi.org/10.3168/jds.2016-11652>.
- Chapin, T. K., K. K. Nightingale, R. W. Worobo, M. Wiedmann, and L. K. Strawn. 2014. Geographical and meteorological factors associated with isolation of *Listeria* species in New York State produce production and natural environments. *J. Food Prot.* 77:1919–1928. <https://doi.org/10.4315/0362-028X.JFP-14-132>.
- Collings, L. K. M., K. L. Proudfoot, and D. M. Veira. 2011. The effects of feeding untreated and formic acid-treated colostrum ad libitum on intake and immunoglobulin levels in dairy calves. *Can. J. Anim. Sci.* 91:55–59. <https://doi.org/10.4141/CJAS10043>.
- Conneely, M., D. P. Berry, J. P. Murphy, I. Lorenz, M. L. Doherty, and E. Kennedy. 2014. Effect of feeding colostrum at different volumes and subsequent number of transition milk feeds on the serum immunoglobulin G concentration and health status of dairy calves. *J. Dairy Sci.* 97:6991–7000. <https://doi.org/10.3168/jds.2013-7494>.
- Cummins, C., D. P. Berry, J. P. Murphy, I. Lorenz, and E. Kennedy. 2017. The effect of colostrum storage conditions on dairy heifer calf serum immunoglobulin G concentration and preweaning health and growth rate. *J. Dairy Sci.* 100:525–535. <https://doi.org/10.3168/jds.2016-10892>.
- Cummins, C., I. Lorenz, and E. Kennedy. 2016. Short communication: The effect of storage conditions over time on bovine colostrum immunoglobulin G concentration, bacteria, and pH. *J. Dairy Sci.* 99:4857–4863. <https://doi.org/10.3168/jds.2015-10276>.
- Cuttance, E. L., W. A. Mason, S. Cranefield, and R. A. Laven. 2025. The effect of three different preservatives on the numbers and types of bacteria, Brix percentage, pH and nutritional composition of bovine colostrum sourced from New Zealand dairy farms. *N. Z. Vet. J.* 73:10–19. <https://doi.org/10.1080/00480169.2024.2392686>.
- Denholm, K. 2022. A review of bovine colostrum preservation techniques. *J. Dairy Res.* 89:345–354. <https://doi.org/10.1017/S0022029922000711>.
- Denholm, K., K. Baxter-Smith, R. Simpson, and P. Williams. 2023. A survey of colostrum management practices on UK dairy farms. *Livestock (Lond.)* 28:150–159. <https://doi.org/10.12968/live.2023.28.4.150>.
- Denholm, K. S., K. Baxter-Smith, A. Haggerty, M. Denholm, P. Williams, and G. Vertenten. 2025. Randomized prospective study exploring health and growth outcomes for 5 days of extended transition milk feeding in pre-weaned Scottish replacement dairy heifers. *Res. Vet. Sci.* 188:105543. <https://doi.org/10.1016/j.rvsc.2025.105543>.
- Denholm, K. S., J. C. Hunnam, E. L. Cuttance, and S. McDougall. 2017. Influence of preservation methods on the quality of colostrum sourced from New Zealand dairy farms. *N. Z. Vet. J.* 65:264–269. <https://doi.org/10.1080/00480169.2017.1342574>.
- EFSA Panel on FEEDAP (European Food Safety Authority Panel on Additives and Products or Substances used in Animal Feed). 2015. Scientific Opinion on the safety and efficacy of sorbic acid and potassium sorbate when used as technological additives for all animal species based on two dossiers from Nutrinova Nutrition Specialties & Food Ingredients GmbH. *EFSA J.* 13:4239. <https://www.efsa.europa.eu/en/efsajournal/pub/4239>.
- Fischer, A. J., Y. Song, Z. He, D. M. Haines, L. L. Guan, and M. A. Steele. 2018. Effect of delaying colostrum feeding on passive transfer and intestinal bacterial colonization in neonatal male Holstein calves. *J. Dairy Sci.* 101:3099–3109. <https://doi.org/10.3168/jds.2017-13397>.
- Frizzo, L. S., L. P. Soto, M. V. Zbrun, E. Bertozzi, G. Sequeira, R. R. Armesto, and M. R. Rosmini. 2010. Lactic acid bacteria to improve growth performance in young calves fed milk replacer and spray-dried whey powder. *Anim. Feed Sci. Technol.* 157:159–167. <https://doi.org/10.1016/j.anifeedsci.2010.03.005>.
- Godden, S. M., J. E. Lombard, and A. R. Woolums. 2019. Colostrum Management for Dairy Calves. *Vet. Clin. North Am. Food Anim. Pract.* 35:535–556. <https://doi.org/10.1016/j.cvfa.2019.07.005>.
- Godden, S. M., E. Royster, J. Timmerman, B. A. Crooker, and N. Brown. 2025. Efficacy of a stabilized liquid potassium sorbate preservative to prevent bacterial proliferation in bovine colostrum and milk. *J. Dairy Sci.* 108:8753–8766. <https://doi.org/10.3168/jds.2025-26478>.
- Godden, S. M., E. Royster, J. Timmerman, and N. O'Sell. 2020. Efficacy of a stabilized liquid potassium sorbate preservative to reduce bacterial proliferation in colostrum or tank milk stored by refrigeration or at room temperature. Page 389 in American Association of Bovine Practitioners Conference Proceedings. VM Publishing Company. <https://doi.org/10.21423/aabppro20208134>.
- Godden, S. M., D. J. Smolenski, M. Donahue, J. M. Oakes, R. Bey, S. Wells, S. Sreevatsan, J. Stabel, and J. Fetrow. 2012. Heat-treated colostrum and reduced morbidity in preweaned dairy calves: results of a randomized trial and examination of mechanisms of effectiveness. *J. Dairy Sci.* 95:4029–4040. <https://doi.org/10.3168/jds.2011-5275>.
- Haggerty, A., C. Mason, K. Ellis, and K. Denholm. 2021. Risk factors for poor colostrum quality and failure of passive transfer in Scottish dairy calves. *J. Dairy Res.* 88:337–342. <https://doi.org/10.1017/S0022029921000686>.
- Haggerty, A., E. Silva, T. Anderson, D. Bell, C. Mason, and K. S. Denholm. 2025. Identifying critical control points for colostrum contamination in first milking colostrum from Scottish dairy herds. *Prev. Vet. Med.* 239:106514. <https://doi.org/10.1016/j.prevetmed.2025.106514>.
- Hill, T. M., H. G. Bateman II, J. M. Aldrich, J. D. Quigley, and R. L. Schlotterbeck. 2013. Evaluation of ad libitum acidified milk replacer programs for dairy calves. *J. Dairy Sci.* 96:3153–3162. <https://doi.org/10.3168/jds.2012-6132>.
- Hyde, R. M., M. J. Green, C. Hudson, and P. M. Down. 2020. Quantitative analysis of colostrum bacteriology on British dairy farms. *Front. Vet. Sci.* 7:601227. <https://doi.org/10.3389/fvets.2020.601227>.
- James, R. E., C. E. Polan, and K. A. Cummins. 1981. Influence of administered indigenous microorganisms on uptake of [iodine-125] gamma-globulin in vivo by intestinal segments of neonatal calves. *J. Dairy Sci.* 64:52–61. [https://doi.org/10.3168/jds.s0022-0302\(81\)82528-3](https://doi.org/10.3168/jds.s0022-0302(81)82528-3).
- Johnson, J. L., S. M. Godden, T. Molitor, T. Ames, and D. Hagman. 2007. Effects of feeding heat-treated colostrum on passive transfer of immune and nutritional parameters in neonatal dairy calves. *J. Dairy Sci.* 90:5189–5198. <https://doi.org/10.3168/jds.2007-0219>.
- Kargar, S., M. Roshan, S. M. Ghoreishi, A. Akhlaghi, M. Kanani, A. R. Abedi Shams-Abadi, and M. H. Ghaffari. 2020. Extended colostrum feeding for 2 weeks improves growth performance and reduces the susceptibility to diarrhea and pneumonia in neonatal Holstein dairy calves. *J. Dairy Sci.* 103:8130–8142. <https://doi.org/10.3168/jds.2020-18355>.
- Mann, S., G. Curone, T. L. Chandler, P. Moroni, J. Cha, R. Bhawal, and S. Zhang. 2020. Heat treatment of bovine colostrum: I. Effects on bacterial and somatic cell counts, immunoglobulin, insulin, and IGF-I concentrations, as well as the colostrum proteome. *J. Dairy Sci.* 103:9368–9383. <https://doi.org/10.3168/jds.2020-18618>.
- Martin, N. H., A. Trmcic, T. H. Hsieh, K. J. Boor, and M. Wiedmann. 2016. The evolving role of coliforms as indicators of unhygienic processing conditions in dairy foods. *Front. Microbiol.* 7:1549. <https://doi.org/10.3389/fmicb.2016.01549>.
- McGuirk, S. M., and M. Collins. 2004. Managing the production, storage, and delivery of colostrum. *Vet. Clin. North Am. Food Anim. Pract.* 20:593–603. <https://doi.org/10.1016/j.cvfa.2004.06.005>.
- Mellado, M., E. Torres, F. G. Veliz, A. de Santiago, U. Macias-Cruz, and J. E. Garcia. 2017. Effect of quality of colostrum on health, growth and immunoglobulin G concentration in Holstein calves in a hot environment. *Anim. Sci. J.* 88:1327–1336. <https://doi.org/10.1111/asj.12773>.
- Moore, D. A., and W. M. Sischo. 2015. Use of Petrifilm for milk or colostrum total plate and coliform counts. Accessed Aug. 10, 2025. <https://s3.wp.wsu.edu/uploads/sites/2147/2015/04/CCAPetriFilmUseFinal.pdf>.
- Morin, M. P., J. Dubuc, P. Freycon, and S. Buczinski. 2021. Short communication: Diagnostic accuracy of the Petrifilm culture system for identifying colostrum with excessive bacterial contamination in

- Quebec dairy herds. *J. Dairy Sci.* 104:4923–4928. <https://doi.org/10.3168/jds.2020-19474>.
- Morrill, K. M., E. Conrad, A. Lago, J. Campbell, J. Quigley, and H. Tyler. 2012. Nationwide evaluation of quality and composition of colostrum on dairy farms in the United States. *J. Dairy Sci.* 95:3997–4005. <https://doi.org/10.3168/jds.2011-5174>.
- Phipps, A. J., D. S. Beggs, A. J. Murray, P. D. Mansell, M. A. Stevenson, and M. F. Pyman. 2016. Survey of bovine colostrum quality and hygiene on northern Victorian dairy farms. *J. Dairy Sci.* 99:8981–8990. <https://doi.org/10.3168/jds.2016-11200>.
- Senyk, G. F., S. M. Kozlowski, P. S. Noar, W. F. Shipe, and D. K. Bandler. 1987. Comparison of dry culture medium and conventional plating techniques for enumeration of bacteria in pasteurized fluid milk. *J. Dairy Sci.* 70:1152–1158. [https://doi.org/10.3168/jds.S0022-0302\(87\)80126-1](https://doi.org/10.3168/jds.S0022-0302(87)80126-1).
- Sofos, J. N., M. D. Pierson, J. C. Blocher, and F. F. Busta. 1986. Mode of action of sorbic acid on bacterial cells and spores. *Int. J. Food Microbiol.* 3:1–17. [https://doi.org/10.1016/0168-1605\(86\)90036-X](https://doi.org/10.1016/0168-1605(86)90036-X).
- Stewart, S., S. Godden, R. Bey, P. Rapnicki, J. Fetrow, R. Farnsworth, M. Scanlon, Y. Arnold, L. Clow, K. Mueller, and C. Ferrouillet. 2005. Preventing bacterial contamination and proliferation during the harvest, storage, and feeding of fresh bovine colostrum. *J. Dairy Sci.* 88:2571–2578. [https://doi.org/10.3168/jds.S0022-0302\(05\)72933-7](https://doi.org/10.3168/jds.S0022-0302(05)72933-7).
- Van Soest, B., M. Weber Nielsen, A. J. Moeser, A. Abuelo, and M. J. VandeHaar. 2022. Transition milk stimulates intestinal development of neonatal Holstein calves. *J. Dairy Sci.* 105:7011–7022. <https://doi.org/10.3168/jds.2021-21723>.
- Westhoff, T. A., and S. Mann. 2025. Effect of frozen storage of bovine colostrum for up to 1 year on concentrations of immunoglobulins and insulin as well as bacterial counts. *JDS Commun.* 6:406–410. <https://doi.org/10.3168/jdsc.2024-0731>.

ORCIDS

Katharine S. Denholm,  <https://orcid.org/0000-0002-5071-6213>

Maria Carvalho,  <https://orcid.org/0009-0008-0436-7930>

Rheinallt Jones  <https://orcid.org/0000-0001-6462-2085>